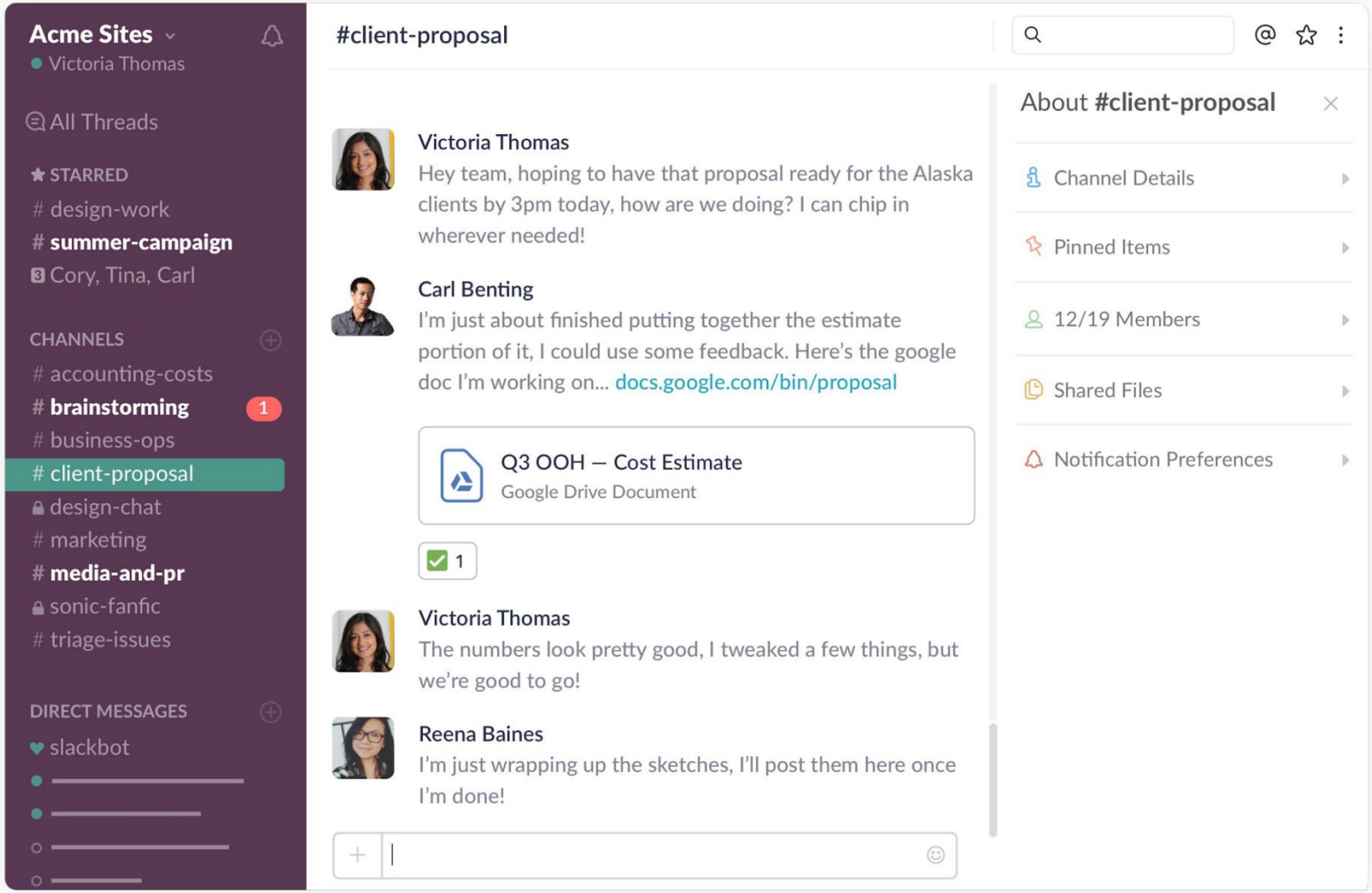
17-423/723: Software System Design

Recitation
Activity: Design for Scalability

Mar 20, 2026

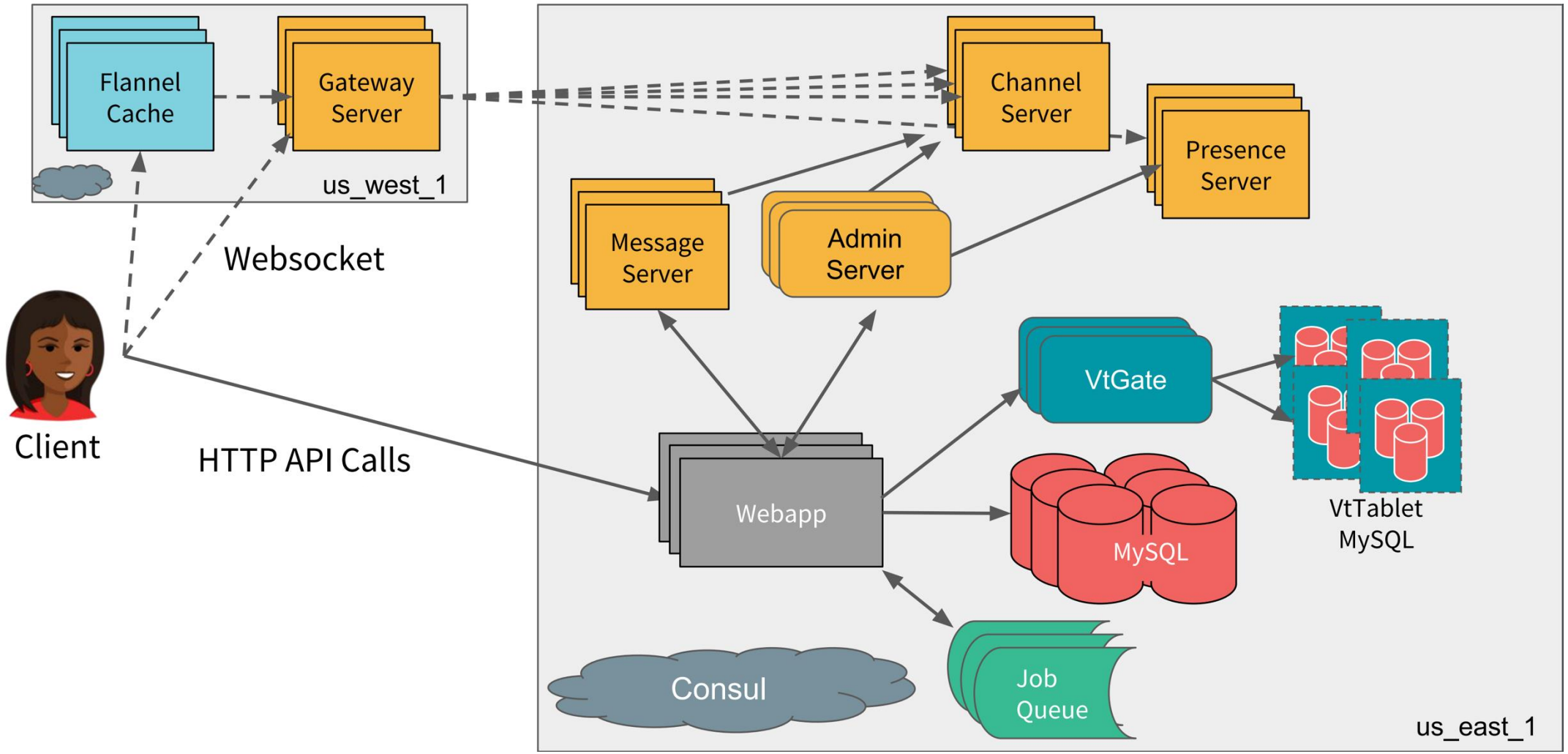
Activity #1: Design a Team Messaging Platform



Activity #1: Design a Team Messaging Platform

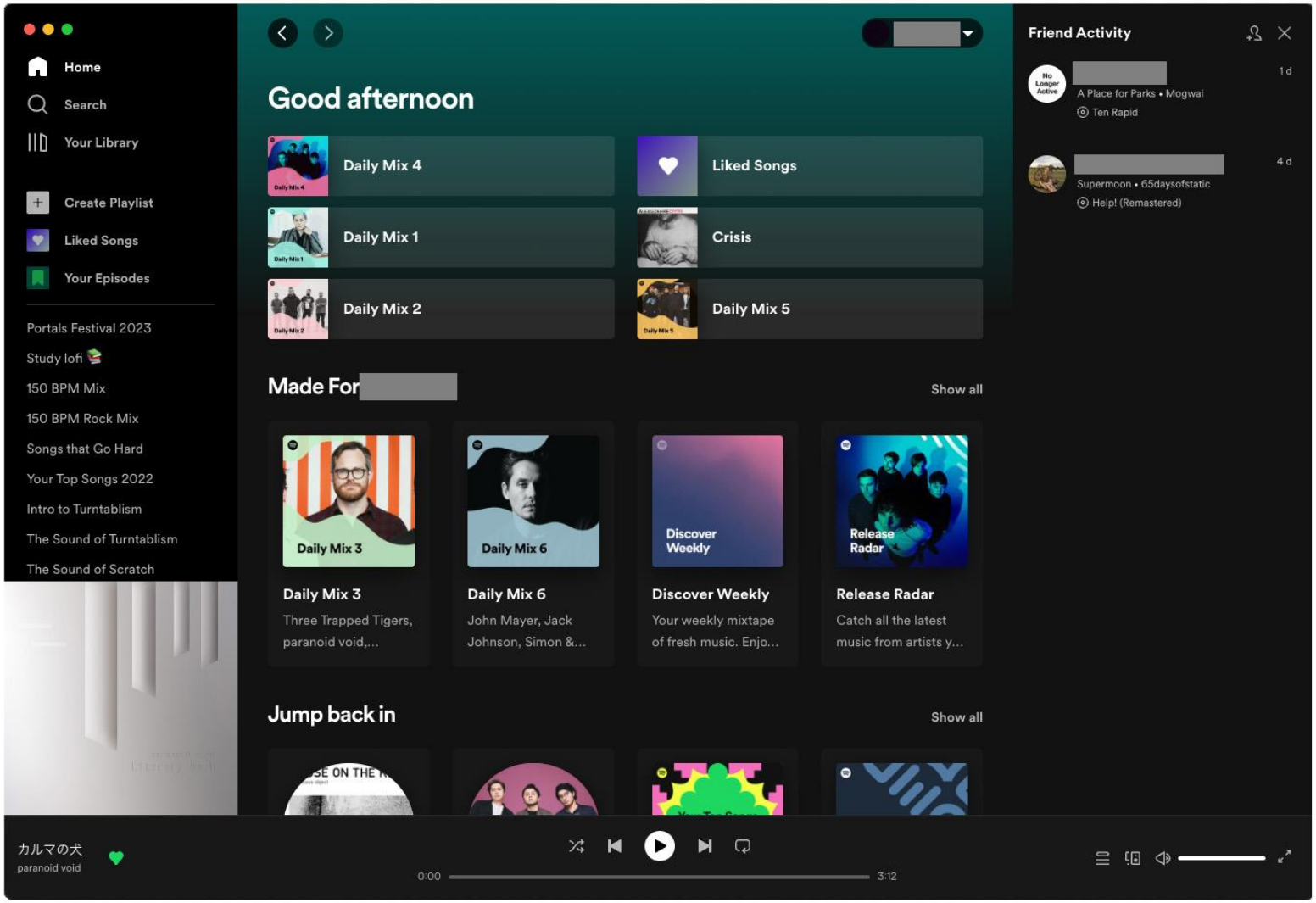
- Sketch a design of a group communication platform (e.g., Slack), focusing on two operations: **Viewing a channel** and **sending a message to a channel**
- **Questions to discuss**:
  1. What data do we need store for the operations?
  2. What types of workload patterns does this app have?
     - Write-heavy vs. ready-heavy? Hot vs. cold data? Steady vs. burst traffic?
     - Where are the likely bottlenecks in the system?
  3. How do we distribute data (replication, partitioning, or both)?
     - If partitioning, how is data partitioned (range-based vs. hashing-based)?
     - If replication, synchronous vs. asynchronous replication?
  4. Which data do we cache? Which cache invalidation policy to use?

Slack Design (2018)



Scaling Slack - The Good, the Unexpected, and the Road Ahead (Mike Demmer, 2018)

Activity #2: Design a Music Streaming App



Activity #2: Design a Music Streaming App

- Sketch a design of a music streaming application (e.g., Spotify), focusing on two operations: **Searching for songs** and **playing a playlist of songs**
- **Questions to discuss**:
  1. What data do we need store for the operations?
  2. What types of workload patterns does this app have?
     - Write-heavy vs. ready-heavy? Hot vs. cold data? Steady vs. burst traffic?
     - Where are the likely bottlenecks in the system?
  3. How do we distribute data (replication, partitioning, or both)?
     - If partitioning, how is data partitioned (range-based vs. hashing-based)?
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  4. Which data do we cache? Which cache invalidation policy to use?